

Environment-Specific Prebiotic Reaction Networks for LIFEORIG

LIFEORIG working model specification

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1 Purpose

The previous PREBIOTIC reference file, `prebiotic_polymer_autocatalytic_network_v2.txt`, is best treated as a master scaffold, not as a single universal chemistry. Early Earth settings differ too strongly in energy source, phase structure, pH, pressure, mineralogy, transport, and concentration mechanism for one reaction list to be used without environmental filtering. I therefore created three sibling environment-specific networks:

Environment	File
Hydrothermal vent	<code>reference_reactions/HYDRO_VENT/hydro_vent_reaction_network_v1.txt</code>
Surface pond	<code>reference_reactions/SURFACE_POND/surface_pond_reaction_network_v1.txt</code>
Volcanic rock	<code>reference_reactions/VOLCANIC_ROCK_ENV/volcanic_rock_reaction_network_v1.txt</code>

These are not validated kinetic mechanisms. They are literature-guided, explicit, environment-conditioned hypothesis networks for simulation and comparison. Each line stores a reaction or lumped pathway, catalyst/control, rate-law template, role, references, and confidence flag.

2 Core Modeling Decision

The correct comparison is not

one generic prebiotic network under different temperatures.

The correct comparison is

a shared reaction library filtered into environment-specific active networks, with different food sets, catalysts, kinetic modifiers, and loss terms.

The overlap is chemically meaningful. For example, `HCN`, formamide, glycolonitrile, aminoacetonitrile, glycine, phosphate species, and short peptides can appear in more than one setting. But the way they are produced, stabilized, destroyed, and concentrated is environment-specific.

3 Environment Distinctions

Feature	Hydrothermal vent	Surface pond	Volcanic rock environment
Dominant energy	Redox disequilibrium, thermal gradients, pH gradients, fluid flow	UV, evaporation, freezing/eutectic concentration, wet-dry cycles	Volcanic heat, UV, gas deposition, wet-dry films, gas-rock redox
Dominant phase	High-pressure aqueous pore/fluid system	Shallow aqueous surface, air-water and ice-water interfaces	Thin films on basalt, glass, ash, clay, sulfides, and pores
Primary catalysts	FeS, NiS, magnetite, brucite, silica, pore barriers	UV, sulfite/sulfide photoredox, clay, borate, phosphate minerals	Basaltic glass, olivine alteration products, clay, FeS, NiS, magnetite, ash surfaces
Concentration mechanism	Pore retention, adsorption, vent-ocean mixing, ice-interface HCN input	Evaporation, freezing, surface enrichment, sediment adsorption	Evaporating rock films, pore trapping, ash/glass adsorption, thermal cycling
Major losses	Advection/dilution, hydrolysis, thermal degradation, oxidant mixing	Photolysis, volatilization, washout, tar formation, precipitation	Volatilization, runoff, thermal degradation, fouling, hydrolysis

4 Hydrothermal Vent Network

The hydrothermal network emphasizes PHREEQC-controlled geochemistry and a separate organic kinetic layer. Acid-base equilibria, carbonate/sulfide/phosphate speciation, mineral saturation, and precipitation should mostly be handled by PHREEQC. The organic layer then uses PHREEQC outputs as activities, pH, ionic strength, redox state, and mineral surface availability.

The vent network has these main modules:

- geochemical speciation and precipitation: carbonate, sulfide, ammonia, phosphate, FeS, NiS, magnetite, brucite, silica;
- serpentinization and water-rock redox: H₂ generation, methanation, magnetite/brucite formation;
- C1 fixation and sulfur carbonyl chemistry: formate, CO, COS, formamide, thioformyl intermediates;
- thioester and acetyl-phosphate activation: thioacetate, acetyl phosphate, acyl transfer, hydrolysis losses;
- central carboxylate and reductive amination chemistry: glyoxylate, pyruvate, oxaloacetate, succinate, amino acids;
- peptide, lipid, mineral-surface, transport, and degradation modules;
- a warm HCN/ice-interface subnetwork from Das, Ghule, and Vanka, included only as a boundary/interface branch rather than as the core vent chemistry.

Important examples include:

```
C02 + H2 -> HC00- + H+
C02 + HS- + H+ -> C0S + H20
C0 + CH3SH -> CH3C0SR
CH3C0SR + Pi -> acetyl_phosphate + RSH
glyoxylate + NH3 + H2 -> glycine + H20
```

5 Surface Pond Network

The surface pond network is the strongest home for HCN, UV, photoredox, wet-dry, and freezing chemistry. It should be used for shallow ponds, lakeshores, tidal lagoons, ice/eutectic brines, and evaporating surface pools. Unlike vents, it explicitly includes UV photolysis and atmospheric deposition as first-class feedstock processes.

Its main modules are:

- atmospheric delivery of HCN, H₂CO, NH₃, SO₂, H₂S, metals, phosphate, and meteoritic organics;
- UV and sulfite/sulfide photoredox chemistry;
- HCN oligomerization, formamide chemistry, nucleobase precursors;
- formose chemistry with borate stabilization;
- Strecker and reductive-amination amino-acid chemistry;
- phosphate activation and wet-dry peptide/oligonucleotide condensation;
- fatty-acid vesicle formation and wet-dry compartment cycling;
- thermal aqueous HCN chemistry from Das, Ghule, and Vanka.

The Das–Ghule–Vanka paper is especially relevant here because it argues that HCN and water alone can generate key RNA/protein precursors under approximately 80–100 °C thermal aqueous conditions, with water or ammonia acting as proton shuttles and without requiring metal catalysts. Reactions added from that subnetwork include:

```
HCN + H2O -> formamide
formamide + H2O -> formic_acid + NH3
formic_acid -> C0 + H20
HCN + H2O -> formaldimine
formaldimine + HCN -> aminoacetonitrile
aminoacetonitrile + 2H2O -> glycine + NH3
urea -> cyanamide + H20
formamide + glycolonitrile -> amino_oxazole_derivative + H20
```

6 Volcanic Rock Network

The volcanic rock network is not simply a dry version of the surface pond. It is centered on rock films, ash beds, basaltic glass, volcanic gases, fumarolic inputs, heat pulses, and mineral surfaces. It includes some HCN and wet-dry chemistry, but the boundary conditions are gas-rock-film and thermal cycling, not open pond photochemistry alone.

Its main modules are:

- volcanic gas input: CO₂, CO, H₂, H₂S, SO₂, COS, HCN, H₂CO, NH₃, NO_x;
- basaltic glass, olivine, ash, clay, apatite, carbonate, and sulfide weathering;
- sulfur-carbonyl and Fe/Ni sulfide surface chemistry;
- C1/C2 reduction and carboxylate formation in thin films;
- HCN/nitrile, formose, Strecker, and thermal HCN-water branches;
- dry heat, wet-dry, mineral-surface peptide and nucleotide polymerization;
- fatty-acid film rehydration, vesicle formation, and pore transport/loss.

Volcanic rock settings therefore bridge atmospheric/surface chemistry and mineral redox chemistry. They can receive atmospheric HCN and formaldehyde, but also offer FeS/NiS, magnetite, basalt glass, phosphate minerals, and dry thermal activation.

7 How To Use These Networks

For computation, the recommended path is:

1. Treat each file as an environment-specific candidate network.
2. Parse reaction IDs, modules, reactants, products, controls, references, and confidence flags.
3. Build a food set from the environment solver: atmosphere, hydrology, PHREEQC, rock weathering, gas fluxes, UV, and mineral availability.
4. Activate reactions only when required feedstocks and environmental controls exist.
5. Use PHREEQC for geochemical equilibria and mineral saturation where flagged; use custom kinetics for organic nonequilibrium pathways.
6. Compare networks by reachable products, autocatalytic closure, feedstock cost, robustness to losses, and overlap with extant metabolic motifs.

8 Immediate Caveats

These networks are intentionally large, but they should not be mistaken for rate-validated mechanisms. The confidence flags are important. High-confidence entries are generally speciation, hydrolysis, adsorption, precipitation, or well-established feedstock transformations. Medium-confidence entries are plausible prebiotic transformations with literature support but uncertain environmental yield. Low-confidence entries are useful hypothesis links, coarse-grained pathways, or autocatalytic/protocell test rules.

9 References

References

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